Programming Fundamentals

Lab 2 Submission

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Program	|BSCS - 1st Semester|
Section	(E)
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Date	|11 Oct 2026|

Lab Task 01

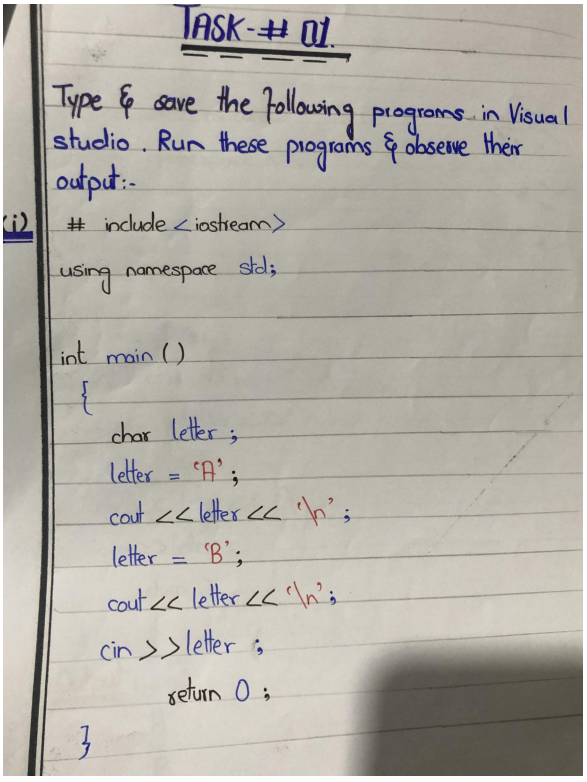
Question:

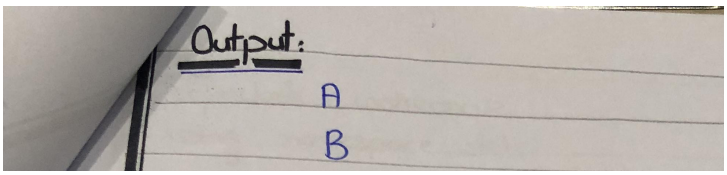
Type and save the following programs, execute them and check the output.

Program A:

```
#include <iostream>
using namespace std;
int main()
{
   char letter;
   letter = 'A';
   cout << letter << "\n";
   letter = 'B';
   cout << letter << "\n";
   cin>>letter;
   return 0;
}
```

Handwritten Code





Code Written On PC

LAB02-TASK01.cpp > main()

```cpp
#include <iostream>
using namespace std;
int main()
{
    char letter;
    letter = 'A';
    cout << letter << "\n";
    letter = 'B';
    cout << letter << "\n";
    cin>>letter;
    return 0;
}
```

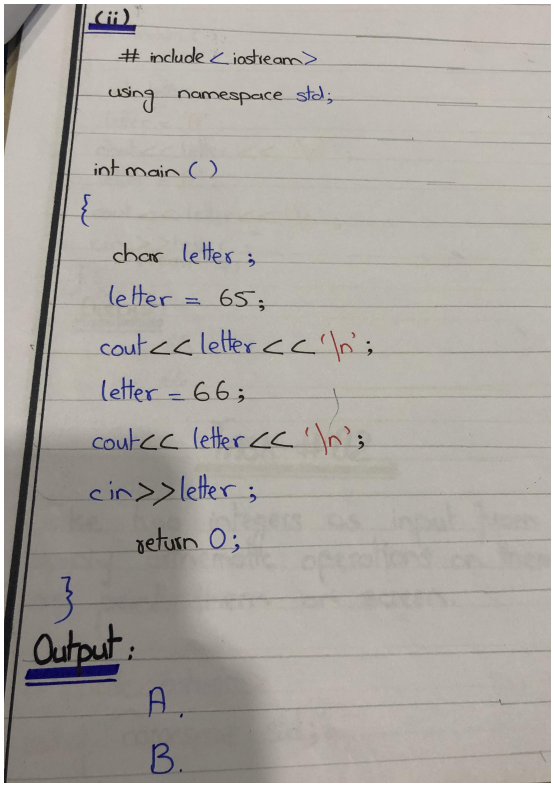
Output

```
[Running] cd "g:\VS Code\hashir-github-demo
Code\hashir-github-demo\"tempCodeRunnerFile
A
B
```

Program B:

```cpp
#include <iostream>
using namespace std;
int main()
{
   char letter;
   letter = 65;
   cout << letter << "\n";
   letter = 66;
   cout << letter << "\n";
   cin>>letter;
   return 0;
}
```

Handwritten code



Code Written On PC

```cpp
#include <iostream>
using namespace std;
int main()
{
    char letter;
    letter = 65;
    cout << letter << "\n";
    letter = 66;
    cout << letter << "\n";
    cin>>letter;
    return 0;
}
```

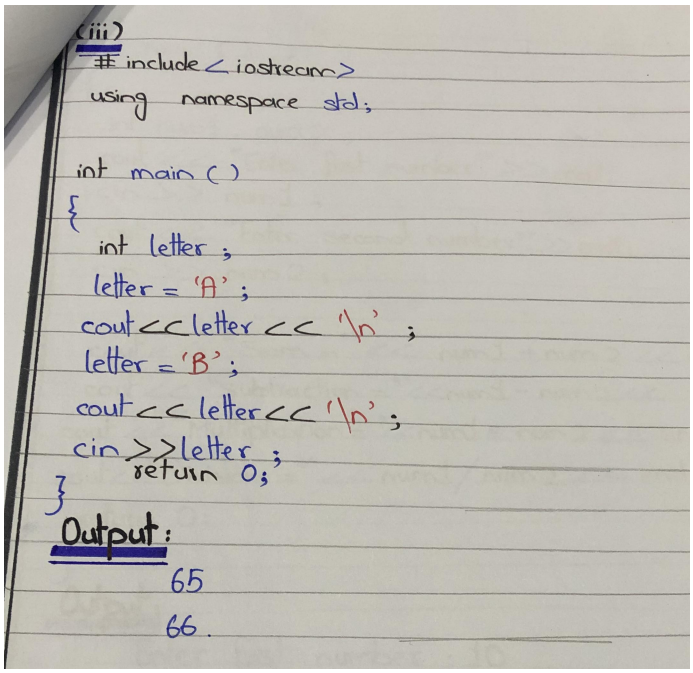
Output

```
[Running] cd "g:\VS Code\hash
Code\hashir-github-demo\"temp
A
B
```

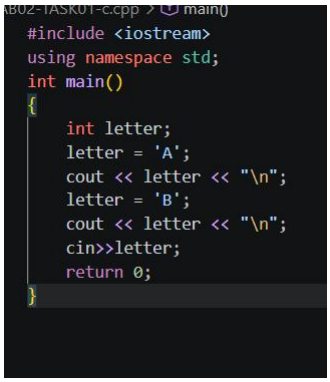
Program C:

```
#include <iostream>
using namespace std;
int main()
{
   int letter;
   letter = 'A';
   cout << letter << "\n";
   letter = 'B';
   cout << letter << "\n";
   cin>>letter;
   return 0;
}
```

Handwritten Code



Code Written On PC



Output

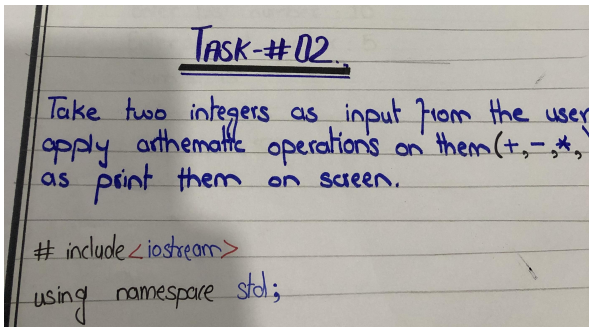
```
[Running] cd "g:\VS Code\hashir-github-der
Code\hashir-github-demo\"LAB02-TASK01-c
65
66
```

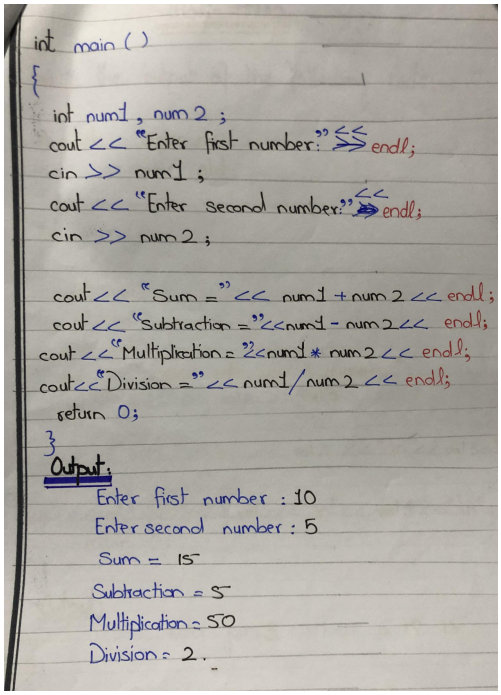
Lab Task 2

Question:

Take two integers as input from the user apply arithmetic operations (+,-,*,/) and print them on the screen.

Handwritten Code





Code Written On PC

```cpp
#include <iostream>
using namespace std;

int main() {
    int num1, num2;

    cout << "Enter first integer: ";
    cin >> num1;

    cout << "Enter second integer: ";
    cin >> num2;

    cout << "\nAddition: " << num1 + num2 << endl;
    cout << "Subtraction: " << num1 - num2 << endl;
    cout << "Multiplication: " << num1 * num2 << endl;

    if (num2 != 0) {
        cout << "Division: " << (double)num1 / num2 << endl;
    } else {
        cout << "Division: Cannot divide by zero!" << endl;
    }

    return 0;
}
```

Output

```
Enter first integer: 10
Enter second integer: 5

Addition: 15
Subtraction: 5
Multiplication: 50
Division: 2

--------------------------------
Process exited after 12.34 seconds with return value 0
Press any key to continue . . .
```

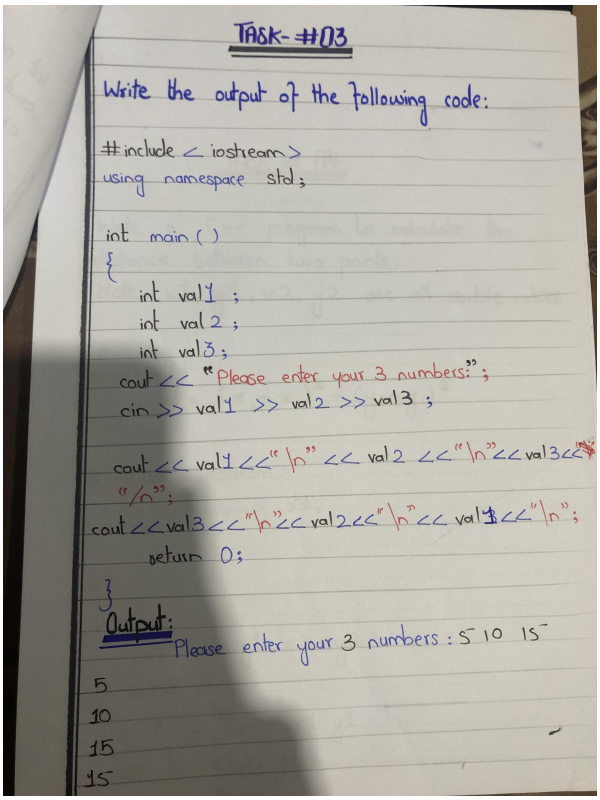
Lab Task 3

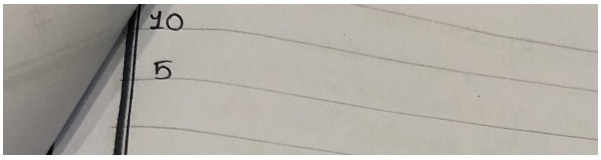
Question:

What will be the output of following code?

```cpp
#include <iostream>
using namespace std;
int main()
{
int val1, val2, val3;
cout << "Please enter your 3 numbers:";
cin >> val1 >> val2 >> val3;
cout << val1 << "\n" << val2 << "\n" << val3 << "\n";
cout << val3 << "\n" << val2 << "\n" << val1 << "\n";
 return 0;
}
```

Handwritten Code





Code Written On PC

```
#include <iostream>
using namespace std;
int main(){
    int val1,val2,val3;
    cout<<"Enter your three numbers:";
    cin>>val1>>val2>>val3;
    cout<<val1<<"\n"<<val2<<"\n"<<val3<<"\n";
    cout<<val3<<"\n"<<val2<<"\n"<<val1<<"\n";
    return 0;

}
```

Output

```
Enter your three numbers:1 2 3
1
2
3
3
2
1

--------------------------------
```

Lab Task 4

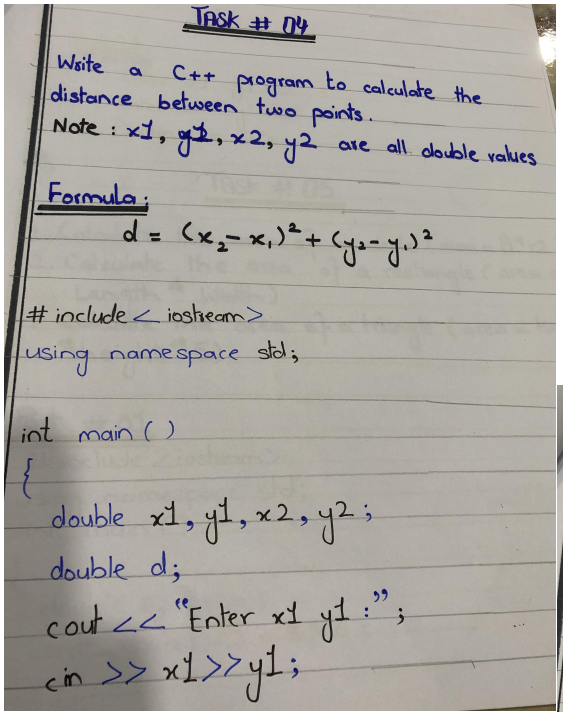
Question:

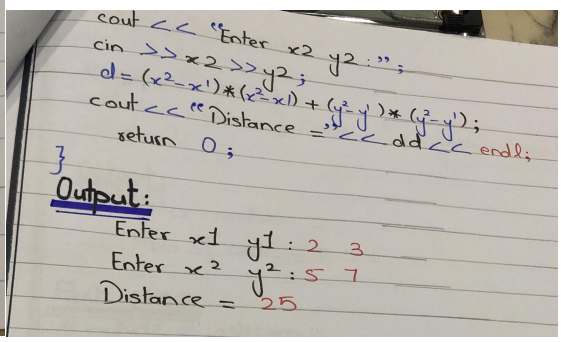
Write a C++ program to calculate distance between two points.

Formula

d = (x2-x1)^2 + (y2-y1)^2. x1,y1,x2,y2 are double.

Handwritten Code





Code Written On PC

```cpp
#include <iostream>
#include <cmath>
using namespace std;

int main() {
    int x1, y1, x2, y2;
    double distance;

    cout << "Enter coordinates of first point (x1, y1): ";
    cin >> x1 >> y1;

    cout << "Enter coordinates of second point (x2, y2): ";
    cin >> x2 >> y2;

    distance = (pow(x2 - x1, 2) + pow(y2 - y1, 2));

    cout << "Distance between two points = " << distance << endl;

    return 0;
}
```

Output

```
Enter coordinates of first point (x1, y1): 2 3
Enter coordinates of second point (x2, y2): 5 7
Distance between two points = 25

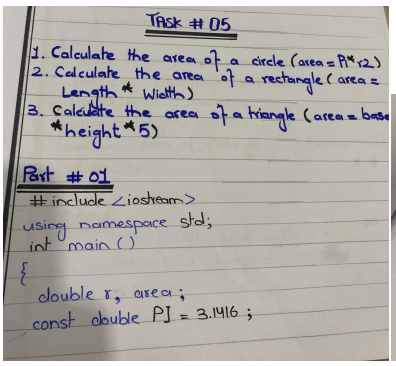
--------------------------------
Process exited after 12 seconds with return value 0
Press any key to continue . . .
```

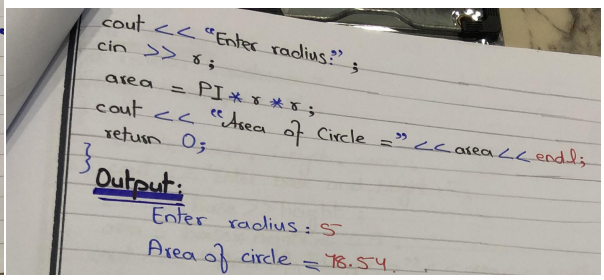
Lab Task 5

Question:

1. Area of Circle PI * r * r
2. Area of Rectangle length * width
3. Area of Triangle base * height * 0.5

NO : 01
Handwritten Code





Code Written On PC

```cpp
#include <iostream>
using namespace std;

int main() {
    float radius, area;
    const float PI = 3.14159;

    cout << "Enter the radius of the circle: ";
    cin >> radius;

    area = PI * radius * radius;

    cout << "Area of circle = " << area << endl;

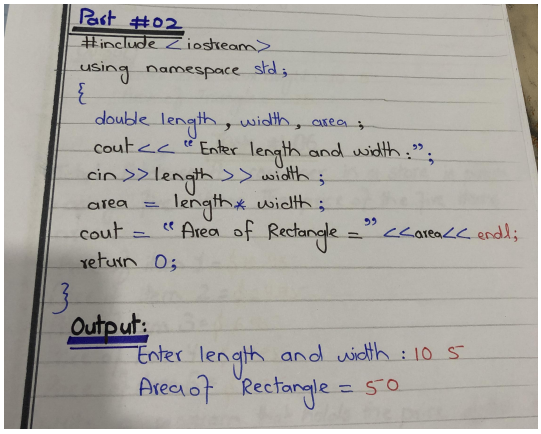
    return 0;
}
```

Output

```
Enter the radius of the circle: 5
Area of circle = 78.5397
```

NO : 02

Handwrittten Code



Code Written On PC

```cpp
#include <iostream>
using namespace std;

int main() {
    double length, width, area;

    cout << "Enter the length and width of the rectangle: ";
    cin >> length>>width;

    area = length*width;

    cout << "Area of rectangle = " << area << endl;

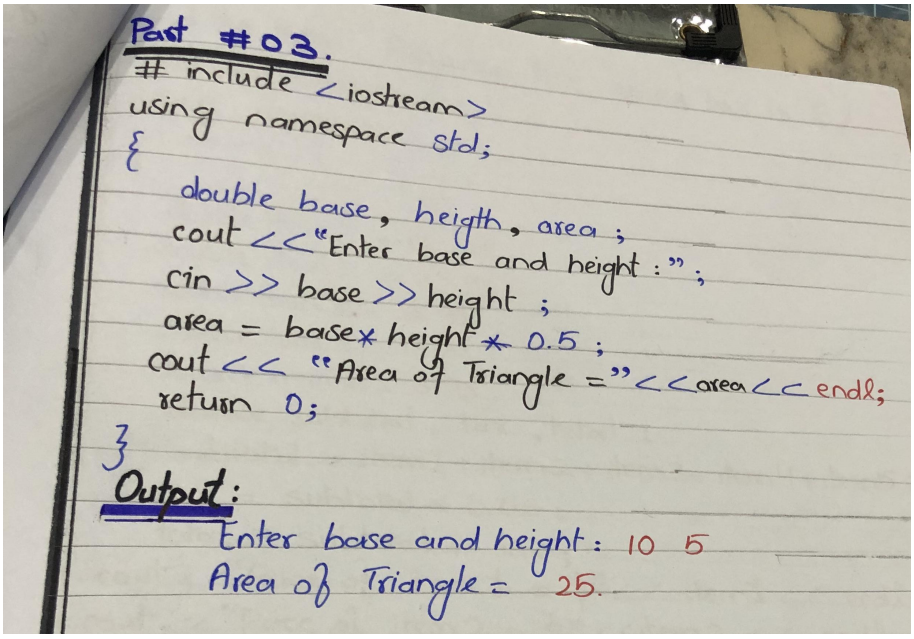
    return 0;
}
```

Output

```
Enter the length and width of the rectangle:  10 5
Area of rectangle = 50
```

NO : 03

Handwritten Code



Code Written On PC

```cpp
#include <iostream>
using namespace std;

int main() {
    double base, height, area;

    cout << "Enter the base and height of the triangle: ";
    cin >> base>>height;
    area = 0.5 * base * height;

    cout << "Area of triangle = " << area << endl;

    return 0;
}
```

Output

```
Enter the base and height of the triangle: 10 5
Area of triangle = 25
```

Lab Task 6

Question:

Total Purchase A customer in a store is purchasing five items. The prices of the five items are
Price Of item 1 =$ 12.95
Price of item 2 =$ 24.95
price of item 3 =$ 6.95
price of item 4 =$ 14.95
Price of item 5 =$ 3.95
Write a program that holds the prices of the five items in five variables. Display each items price, the subtotal of the sale, the amount of sales tax, and the total. Assume the sales tax is 6%

Code Written On PC

```cpp
#include <iostream>
using namespace std;

int main()
{
    double item1 = 12.95;
    double item2 = 24.95;
    double item3 = 6.95;
    double item4 = 14.95;
    double item5 = 3.95;

    double subtotal, tax, total;

    subtotal = item1 + item2 + item3 + item4 + item5;

    tax = subtotal * 0.06;
    total = subtotal + tax;

    cout << "Price of item 1: " << item1 << endl;
    cout << "Price of item 2: " << item2 << endl;
    cout << "Price of item 3: " << item3 << endl;
    cout << "Price of item 4: " << item4 << endl;
    cout << "Price of item 5: " << item5 << endl;

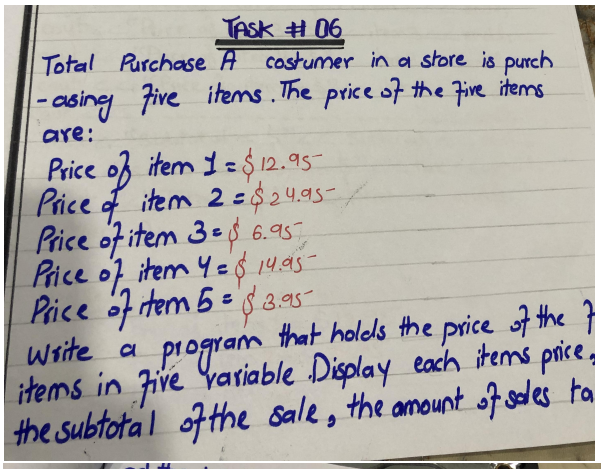
    cout << "Subtotal: " << subtotal << endl;
    cout << "Sales tax (6%): " << tax << endl;
    cout << "Total: " << total << endl;

    return 0;
}
```

Output

```
Price of item 1: 12.95
Price of item 2: 24.95
Price of item 3: 6.95
Price of item 4: 14.95
Price of item 5: 3.95
Subtotal: 63.75
Sales tax (6%): 3.825
Total: 67.575
```

Handwritten Code

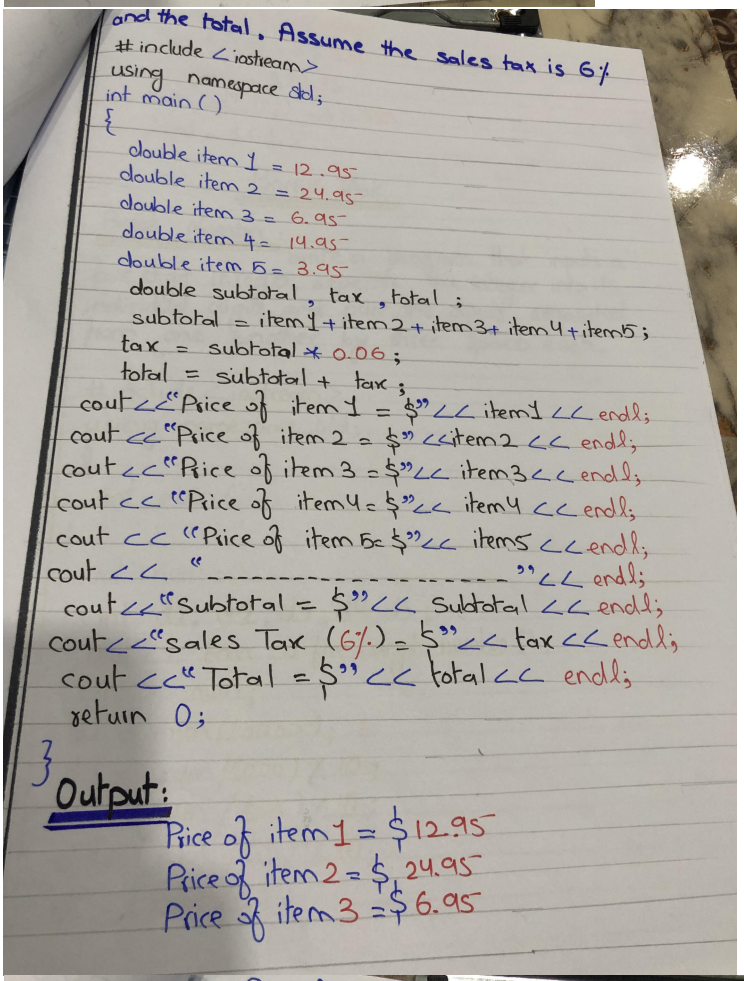


TASK # 06

Total Purchase A costumer in a store is purch -asing five items. The price of the five items are:

Price of item 1 = $12.95
Price of item 2 = $24.95
Price of item 3 = $6.95
Price of item 4 = $14.95
Price of item 5 = $3.95

Write a program that holds the price of the 5 items in five variable. Display each items price, the subtotal of the sale, the amount of sales ta

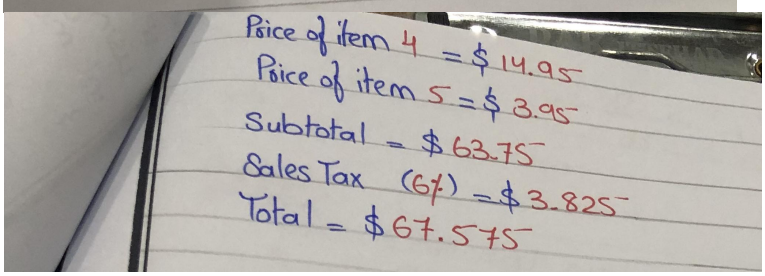


and the total. Assume the sales tax is 6%

```
#include <iostream>
using namespace std;
int main()
{
    double item1 = 12.95;
    double item2 = 24.95;
    double item3 = 6.95;
    double item4 = 14.95;
    double item5 = 3.95;
    double subtotal, tax, total;
    subtotal = item1 + item2 + item3 + item4 + item5;
    tax = subtotal * 0.06;
    total = subtotal + tax;
cout << "Price of item 1 = $" << item1 << endl;
cout << "Price of item 2 = $" << item2 << endl;
cout << "Price of item 3 = $" << item3 << endl;
cout << "Price of item 4 = $" << item4 << endl;
cout << "Price of item 5 = $" << item5 << endl;
cout << "-------------------" << endl;
cout << "subtotal = $" << subtotal << endl;
cout << "sales Tax (6%) = $" << tax << endl;
cout << "Total = $" << total << endl;
return 0;
}
```

Output:

```
Price of item 1 = $12.95
Price of item 2 = $24.95
Price of item 3 = $6.95
```



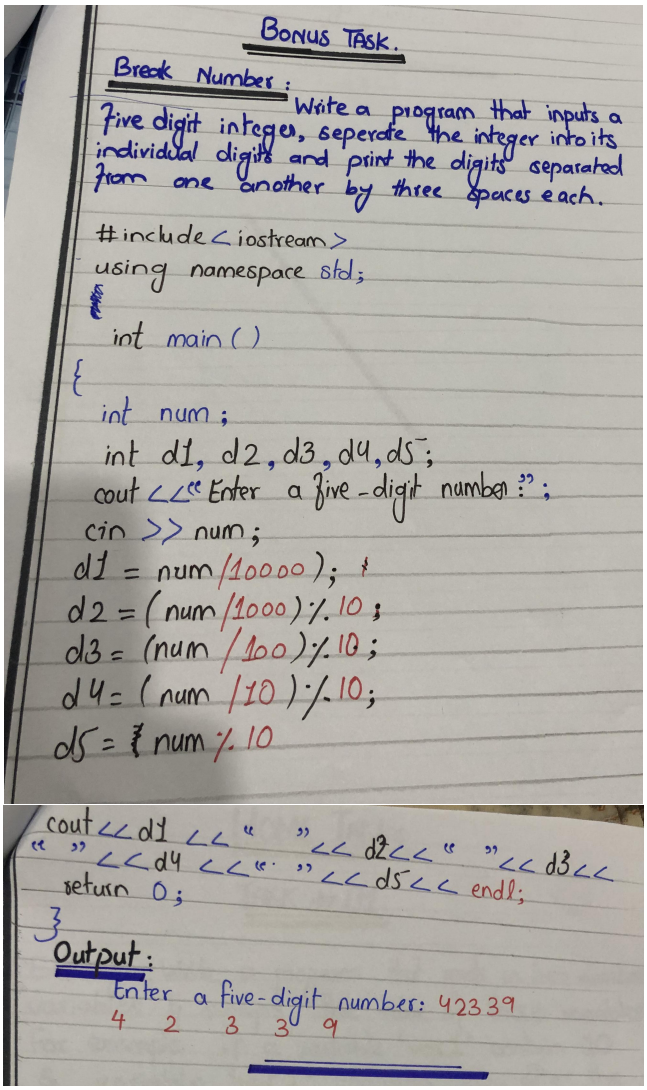
```
Price of item 4 = $14.95
Price of item 5 = $3.95
Subtotal = $63.75
Sales Tax (6%) = $3.825
Total = $67.575
```

Bonus Task

Question:

Write a program to input a five-digit integer and separate into individual digits with 3 spaces. Hint: use division and modulus %.

Handwritten Code



Code Written On PC

```cpp
#include <iostream>
using namespace std;

int main()
{
    int num;
    int d1, d2, d3, d4, d5;

    cout << "Enter a five-digit number: ";
    cin >> num;

    d1 = num / 10000;
    d2 = (num / 1000) % 10;
    d3 = (num / 100) % 10;
    d4 = (num / 10) % 10;
    d5 = num % 10;

    cout << d1 << " " << d2 << " " << d3 << " "
         << d4 << " " << d5 << endl;

    return 0;
}
```

Output

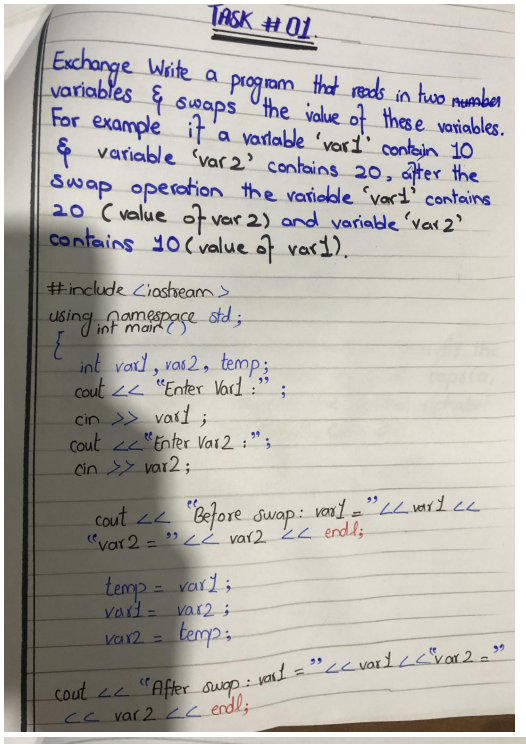
```
Enter a five-digit number: 42339
4 2 3 3 9
```

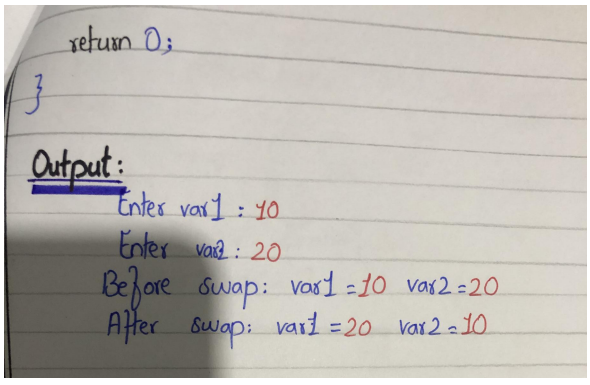
Home Task 1

Question:

Exchange Write a program that reads in two variables and swaps the value of these variables. For example if a variable 'varl' contains 10 and variable 'var2' contains 20, after the swap operation the variable 'varl' contains 20 (value of var2) and variable 'var2' contains 10 (value of varl)

Handwritten Code





Code Written On PC

```cpp
#include <iostream>
using namespace std;

int main()
{
    int var1, var2, temp;

    cout << "Enter var1: ";
    cin >> var1;

    cout << "Enter var2: ";
    cin >> var2;

    cout << "Before swap: var1 = " << var1
         << ", var2 = " << var2 << endl;

    temp = var1;
    var1 = var2;
    var2 = temp;

    cout << "After swap: var1 = " << var1
         << ", var2 = " << var2 << endl;

    return 0;
}
```

Output

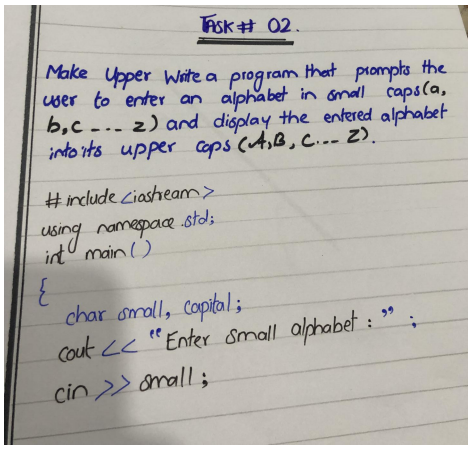
```
Enter var1: 10
Enter var2: 20
Before swap: var1 = 10, var2 = 20
After swap: var1 = 20, var2 = 10
```

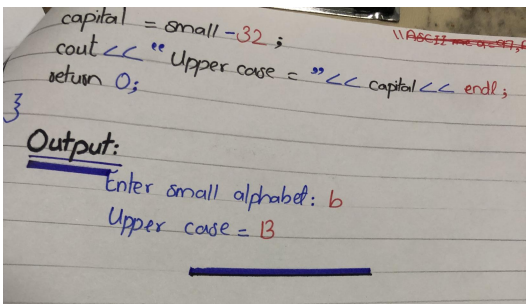
Home Task 2

Question:

Make Upper Write a program that prompts the user to enter an alphabet in small caps (a, b,c . . . z) and display the entered alphabet into its upper caps (A, B, C . . .Z)

Handwritten Code





Code Written On PC

```cpp
#include <iostream>
using namespace std;

int main()
{
    char small, capital;

    cout << "Enter small alphabet: ";
    cin >> small;

    capital = small - 32;

    cout << "Upper case = " << capital << endl;

    return 0;
}
```

Output

```
Enter small alphabet: b
Upper case = B
```
